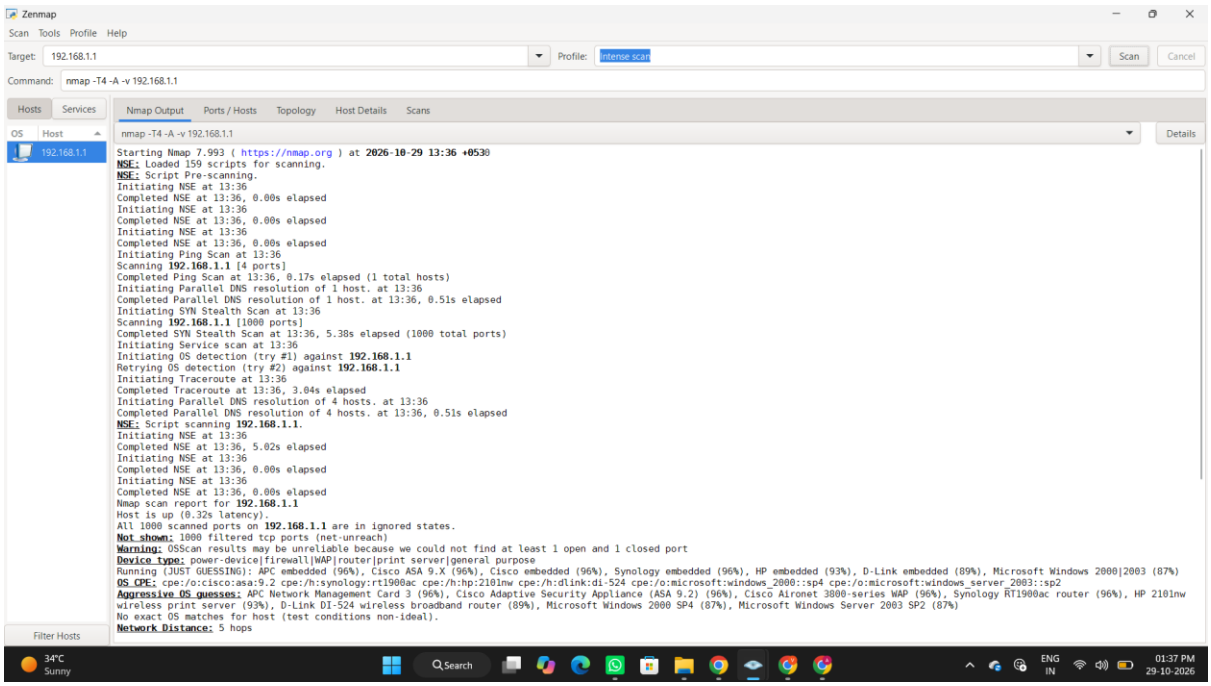
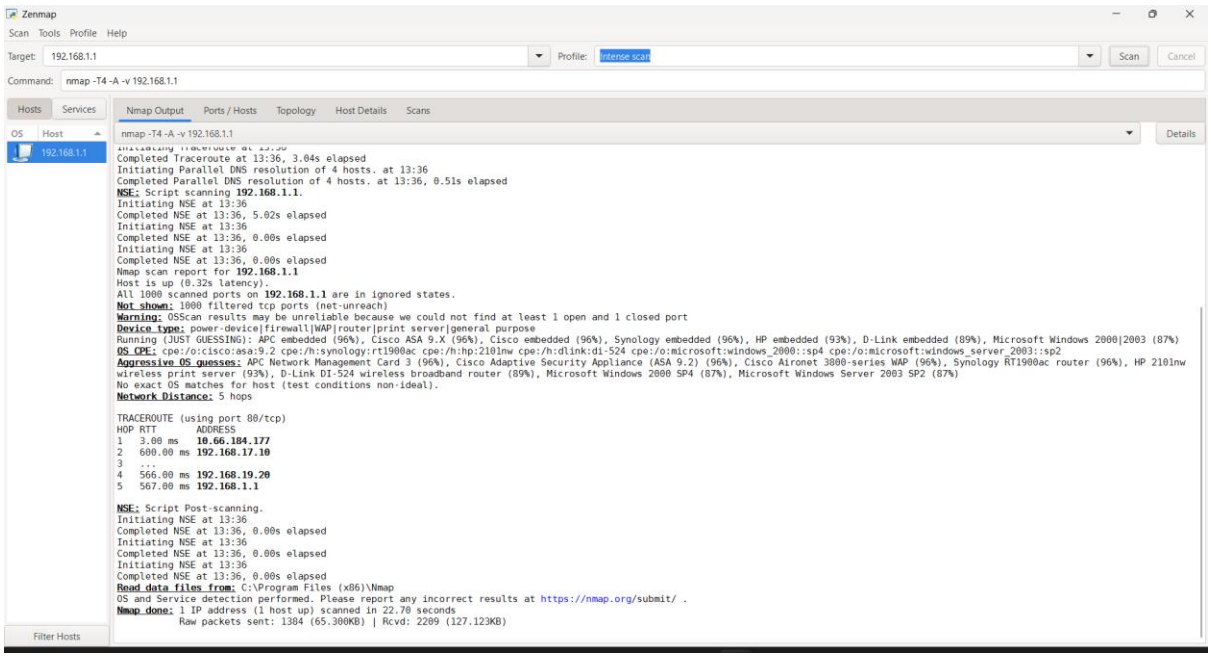
1) **nmap 192.168.1.1→Scan common ports**

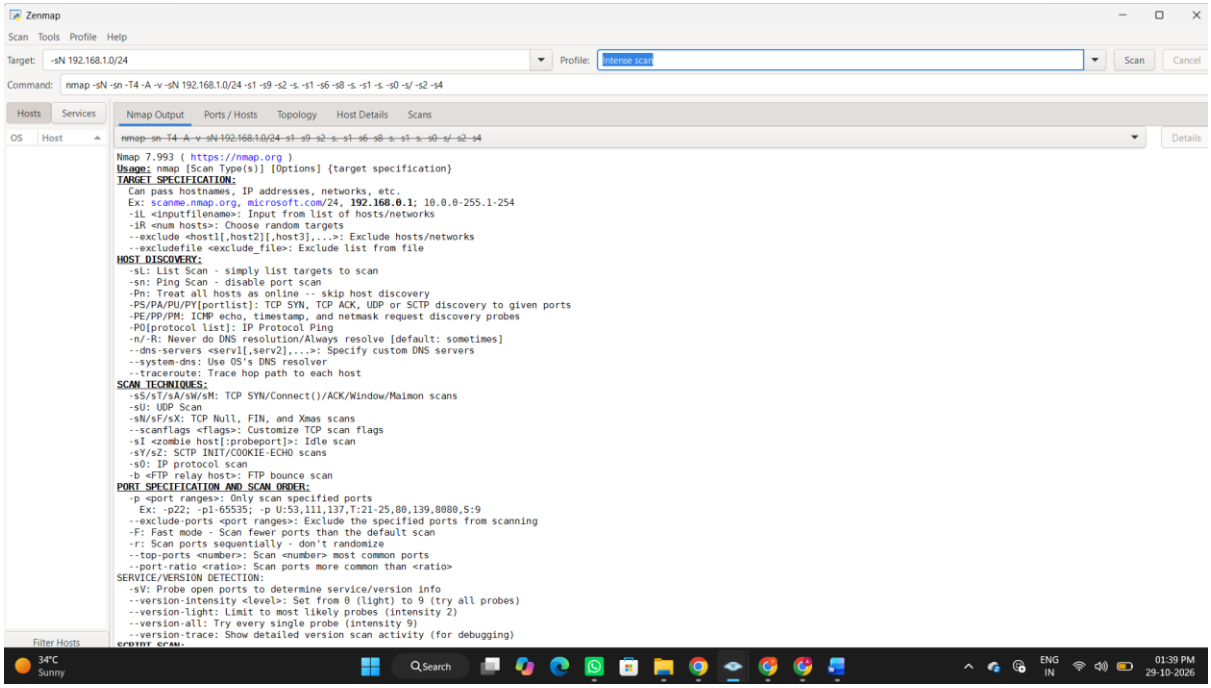
Purpose with Simple Example: Scans common ports to check which ports are open on a router or target device.

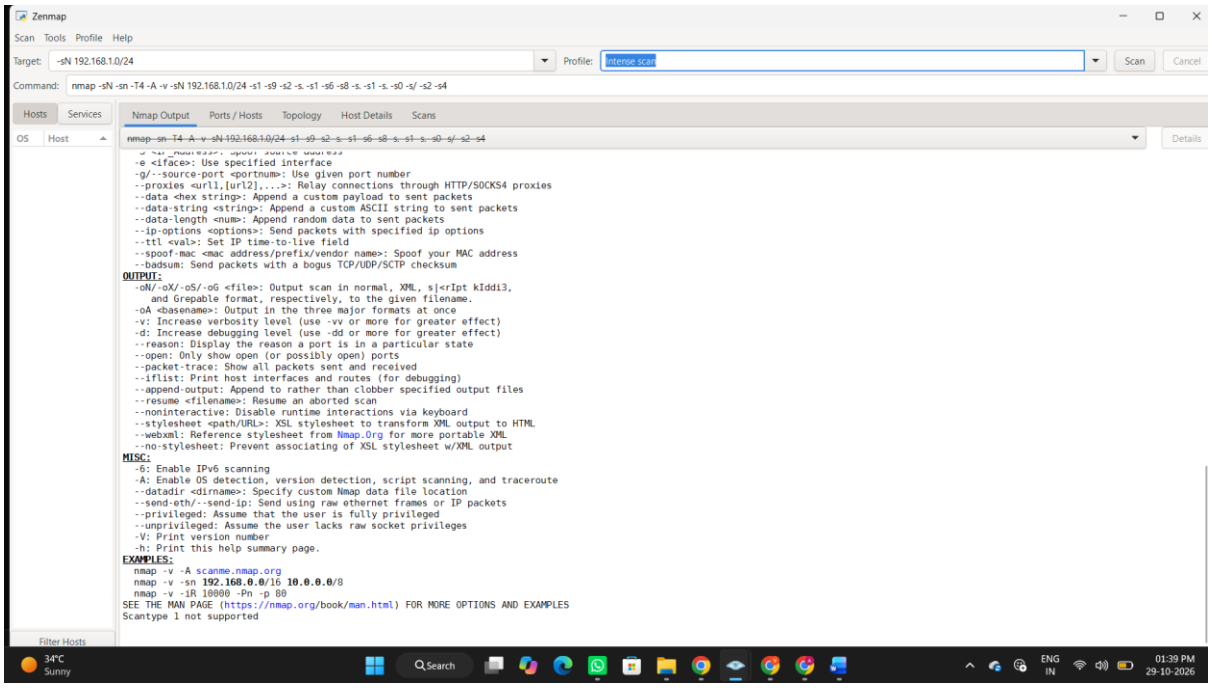




2) **nmap -sn 192.168.1.0/24 → Discover live hosts**

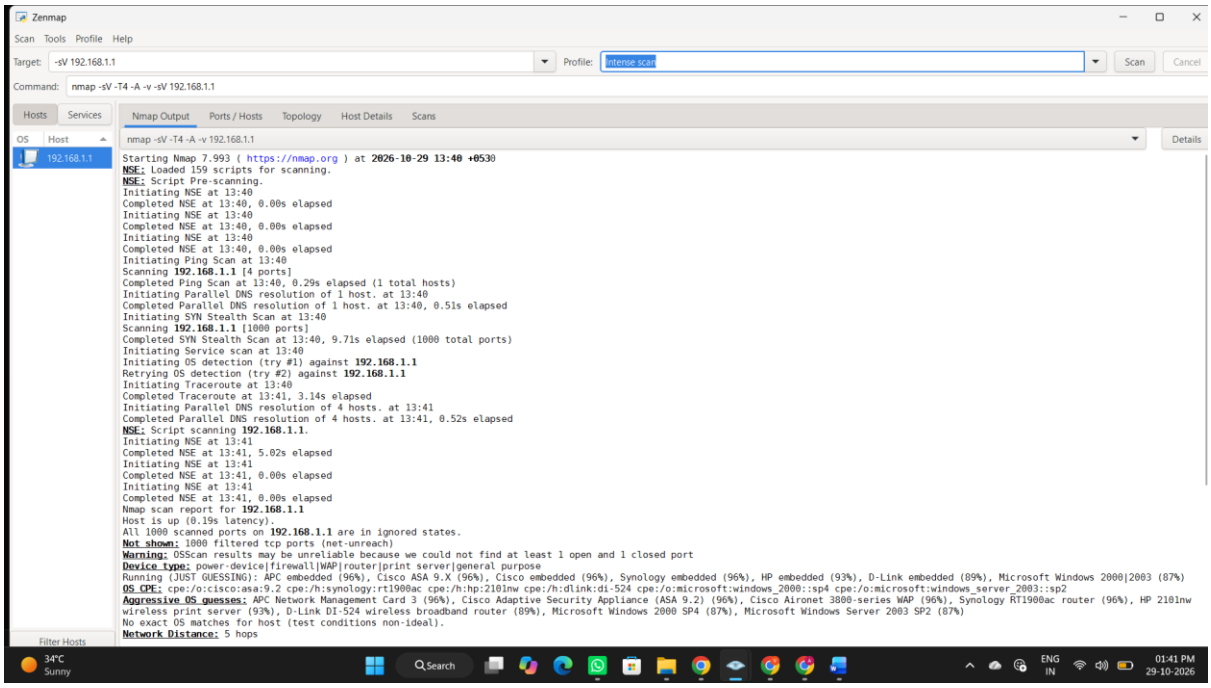
Purpose with Simple Example: Discovers active devices on a network. For example, it can find devices connected to a Wi-Fi network.

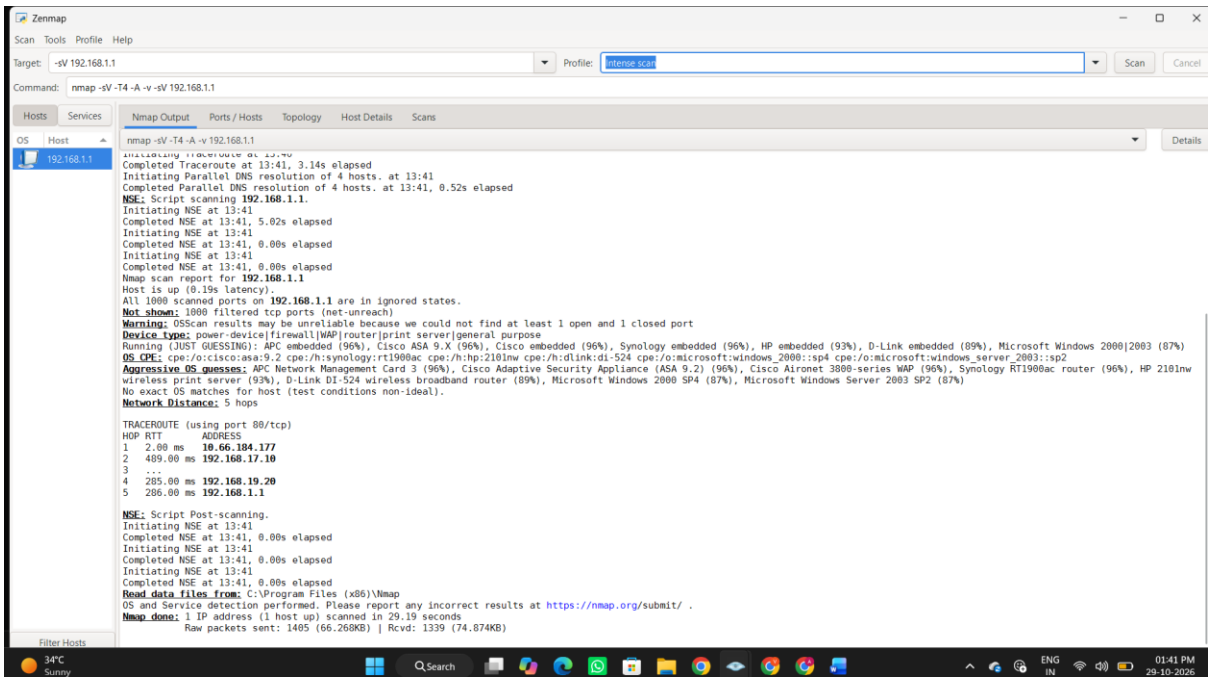




3) **nmap -sV 192.168.1.1 → Detect service versions**

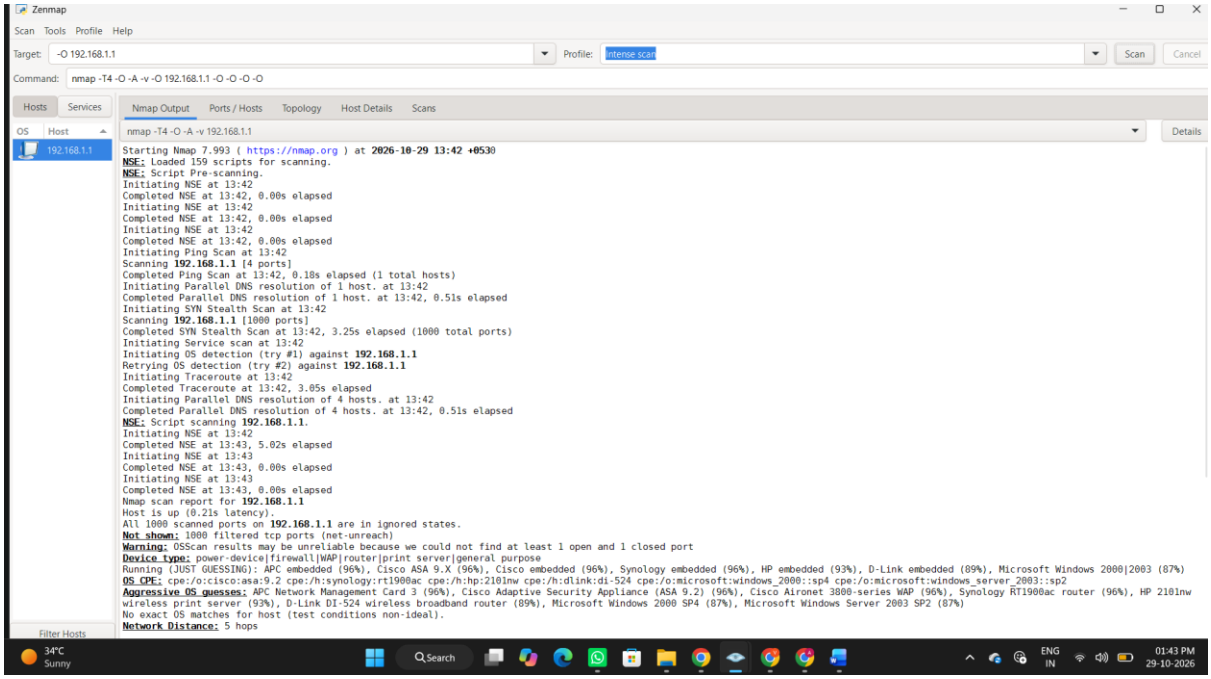
Purpose with Simple Example: Detects service versions running on open ports. For example, it may identify the version of a web server.

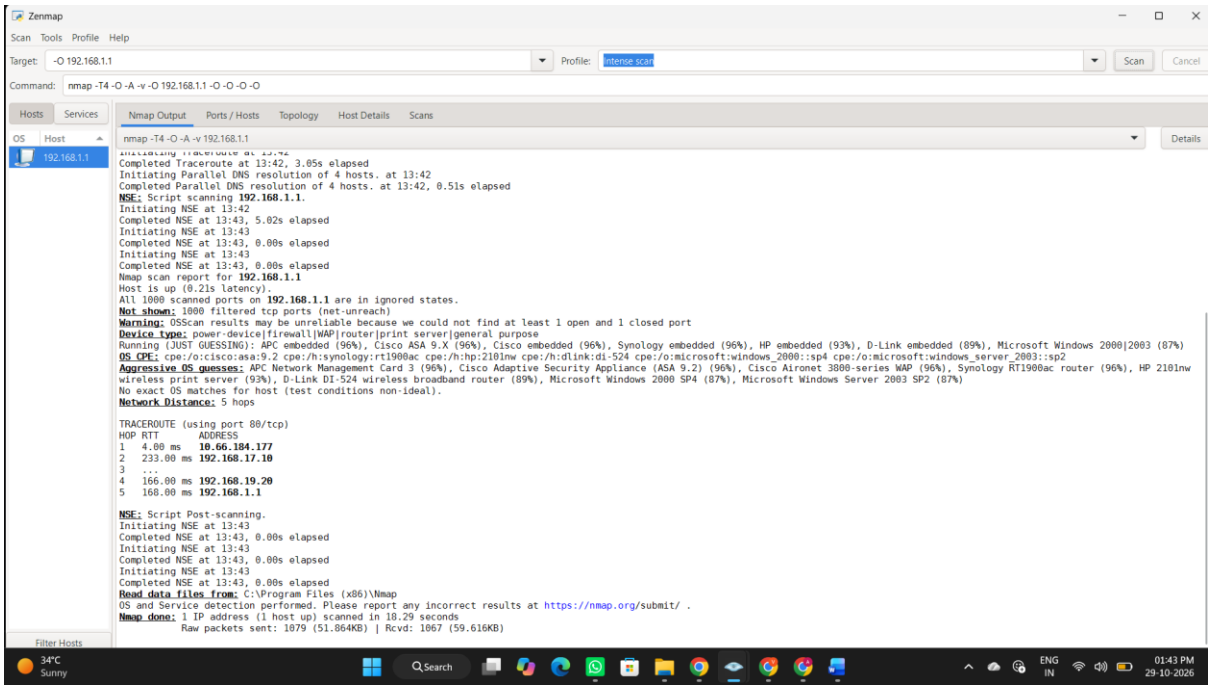




4) nmap -O 192.168.1.1 → Identify the operating system

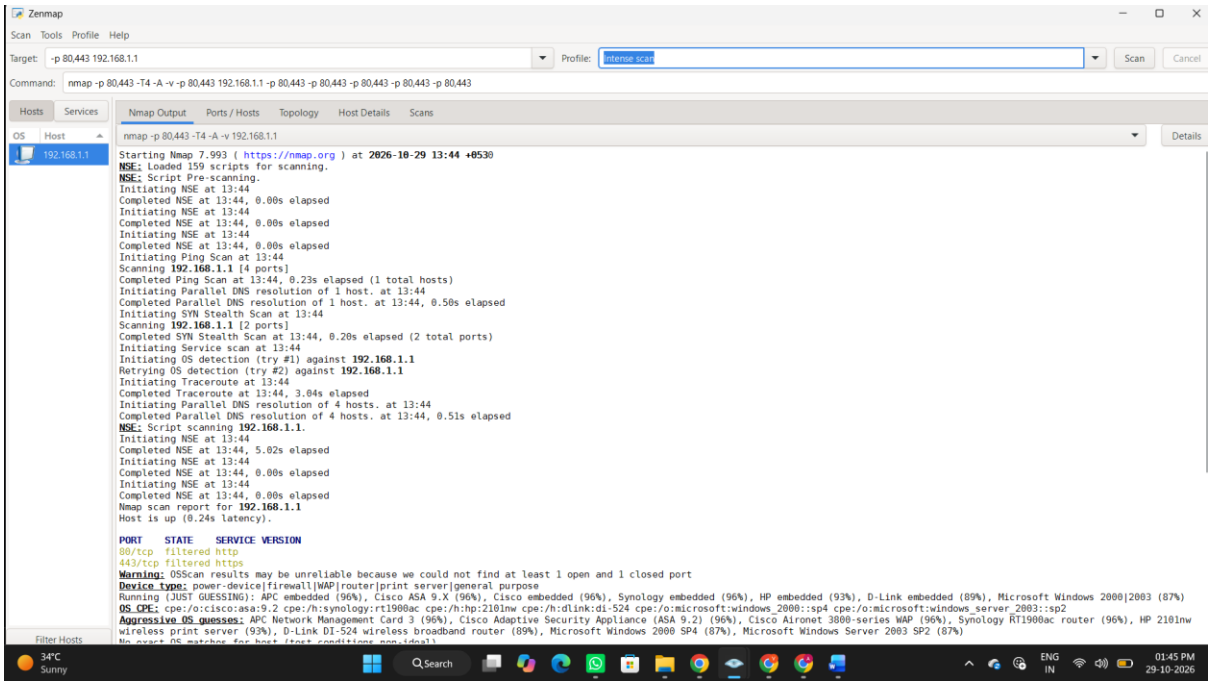
Purpose with Simple Example: Tries to identify the target device's operating system, such as Windows or Linux.

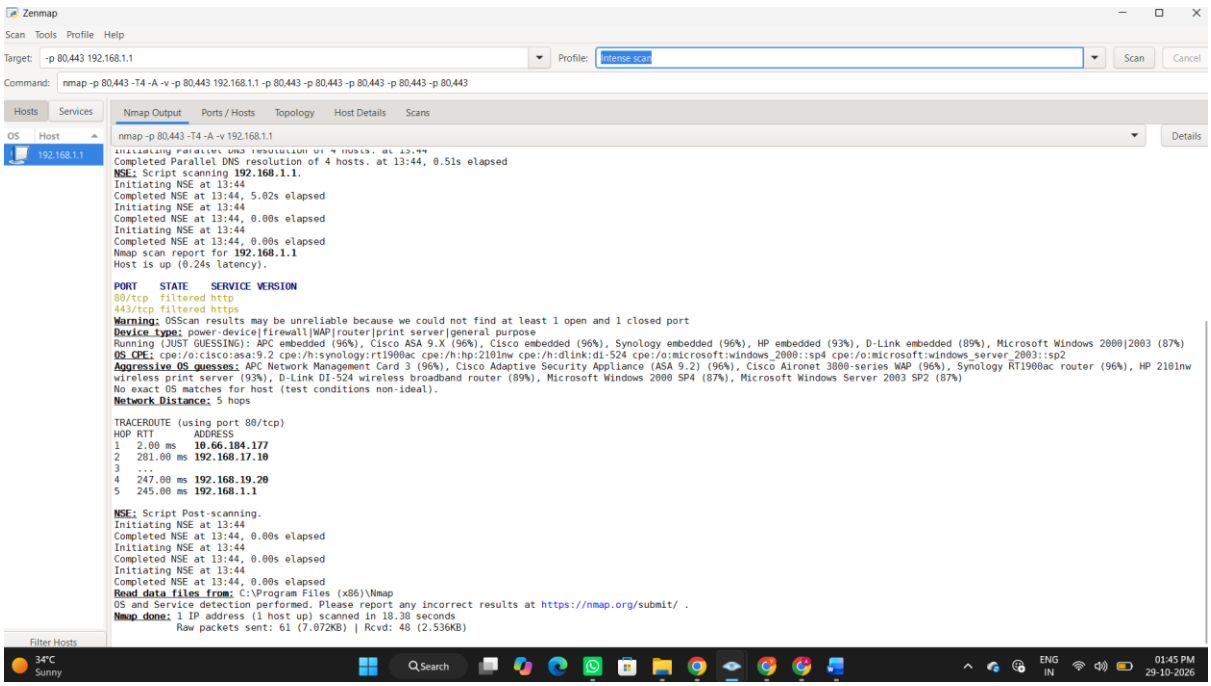




5) nmap -p 80,443 192.168.1.1 → Scan specific ports

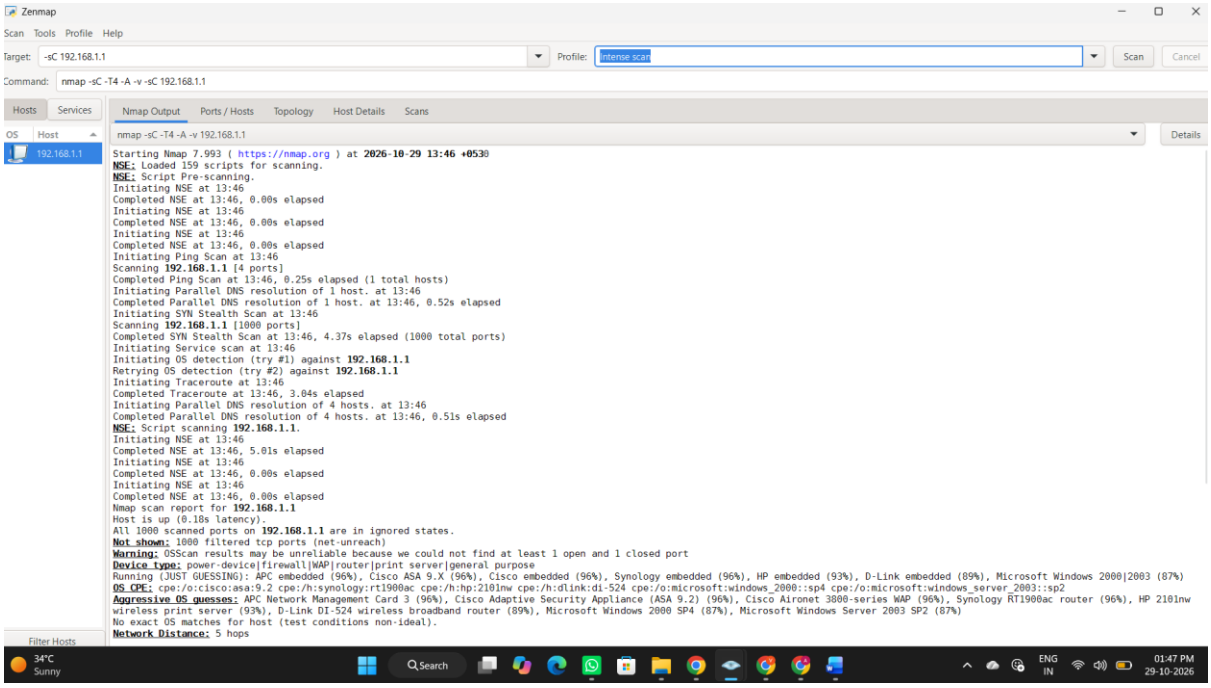
Purpose with Simple Example: Scans specific ports to check whether HTTP (port 80) and HTTPS (port 443) are open.

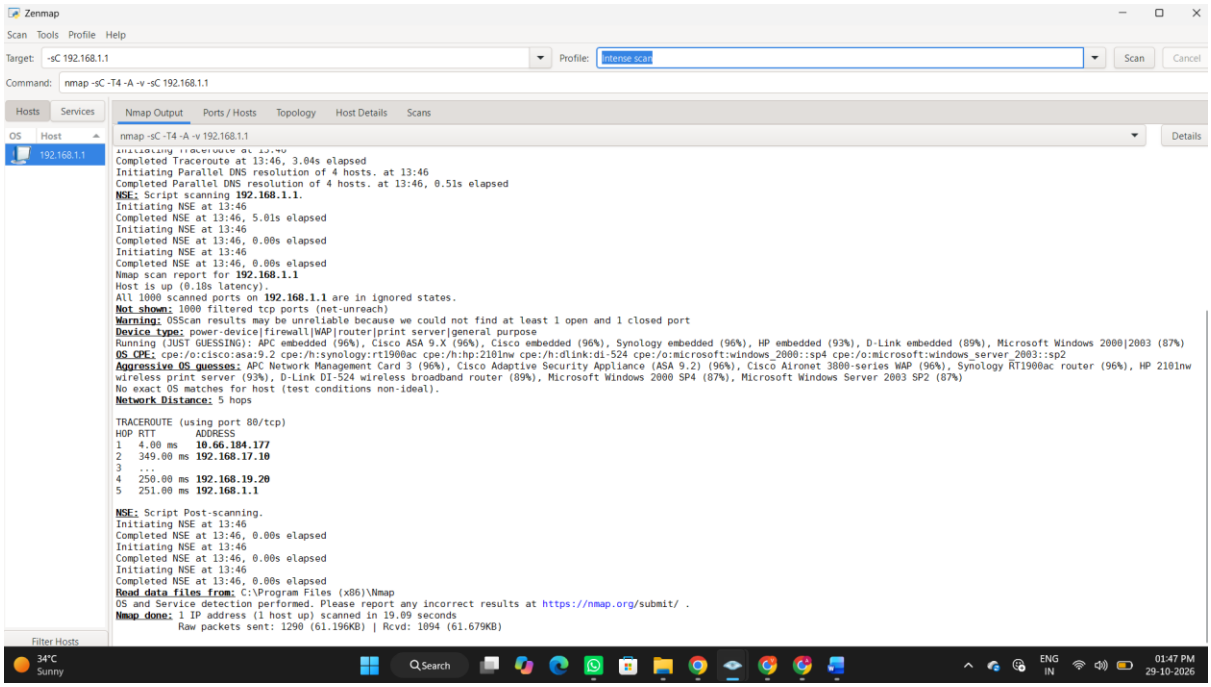




6) nmap -sC 192.168.1.1 → Run default NSE scripts

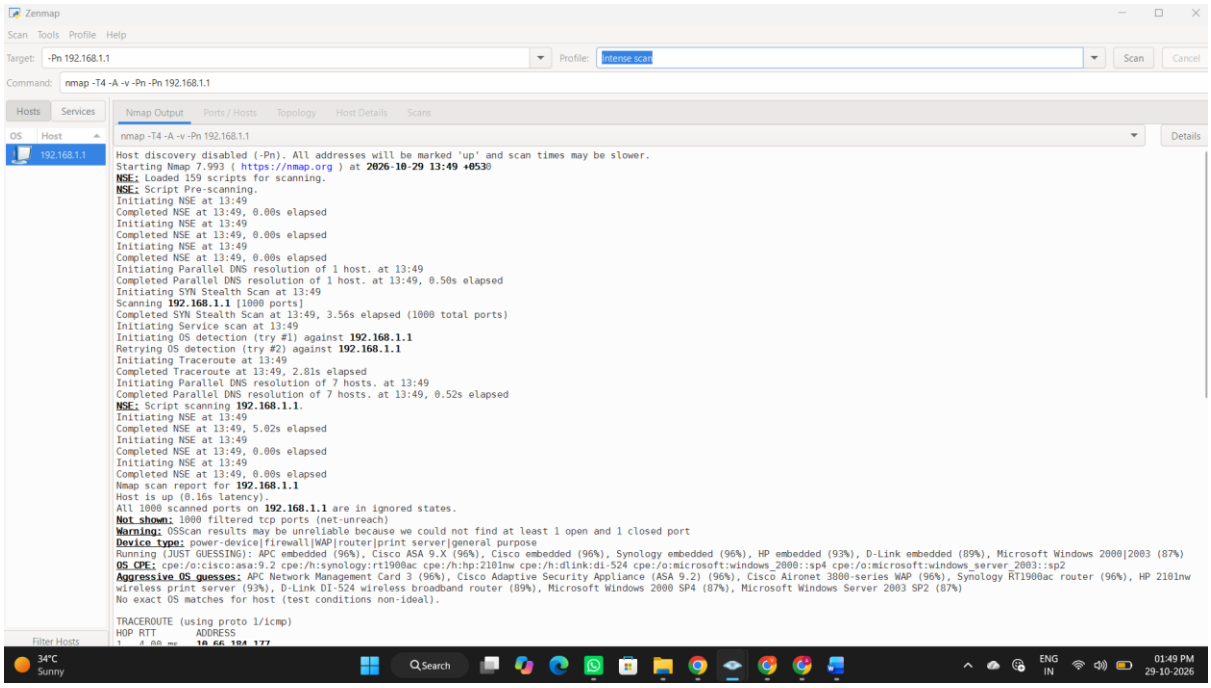
Purpose with Simple Example: Runs default scripts to collect additional information about the target device.

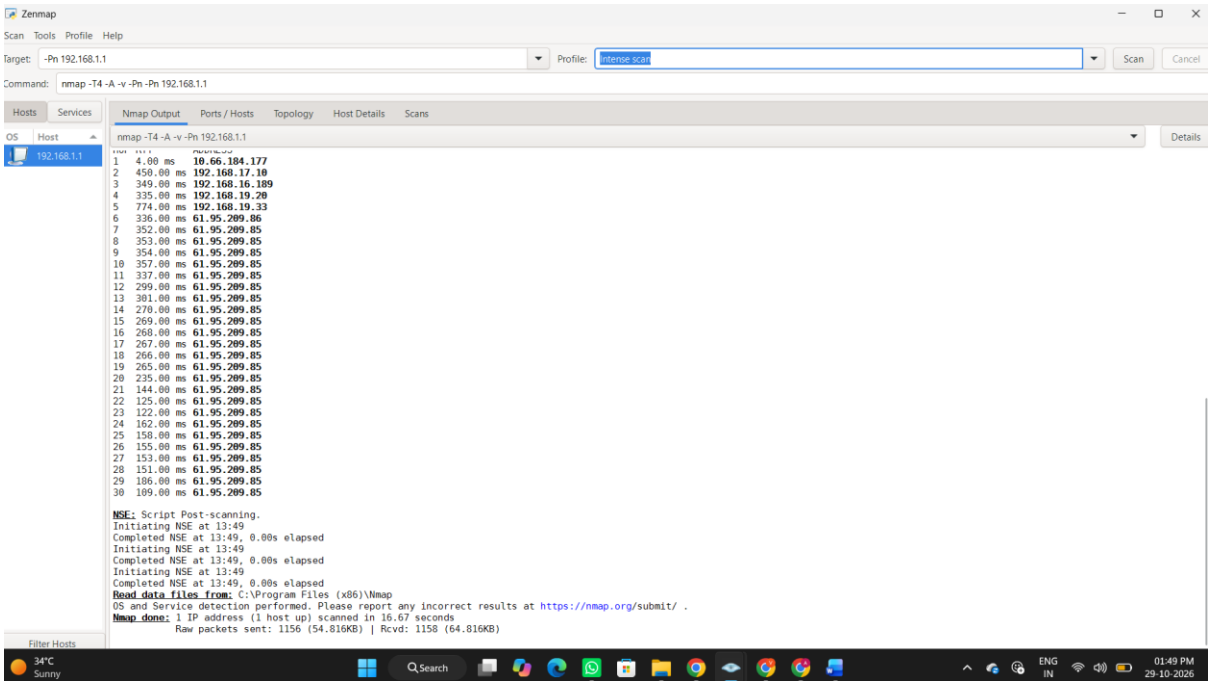




7) nmap -Pn 192.168.1.1 → Skip host discovery

Purpose with Simple Example: Skips host discovery and scans the target directly, even if it does not respond to discovery probes.





8) nmap -oN scan.txt 192.168.1.1 → Save scan results to a file

Purpose with Simple Example: Saves the scan results in a text file named scan.txt for future reference.

